

# 11. Generating an Evidence-Backed Report

Assemble verified records, draft complete sections and check the report before validator approval.

# Move from finding records to a reviewable report

## EVIDENCE STORE

Ingest checked results and source links.

Keep methods, conditions and limitations together.

## FIELD ASSEMBLY

Query the approved report schema.

Fill supported fields and identify evidence gaps.

## DRAFT AND VERIFY

Write from the assembled records.

Check claims, required coverage and cross-section consistency.

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**Software handles records and checks; the validator assesses interpretation, materiality and final approval.**

# Declare what each section must contain

## REGIONAL FINDING SECTION

Model and test version.  
Selected membership rule.  
Sample, conditions and support.  
Contrast, coverage and uncertainty.

## REQUIRED QUALIFICATIONS

How the region was selected.  
Whether confirmation was independent.  
Population and evidence limits.  
Any separate repair assessment.

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A schema makes required evidence and material omissions visible before drafting.

# Build source-linked reporting records

## INGEST

Read run outputs,  
approved notes and  
source tables.

Propose typed assertions.

## CHECK

Verify values, units,  
sample IDs and scope.

Resolve or flag extraction  
conflicts.

## STORE

Link each result to its test,  
conditions and source  
address.

Retain original evidence.

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**Exact reporting values come from checked records with versioned provenance.**

# Query the schema into concrete fields

| Required field | Evidence request                                  |
|----------------|---------------------------------------------------|
| Method         | Which model, representation and selection rule?   |
| Effect         | What region/rest loss and contrast were measured? |
| Support        | How many cases, and what affected share?          |
| Uncertainty    | Which interval and multiplicity contract?         |
| Scope          | Which population and unresolved limitations?      |

**Keep query, returned record IDs and missing-field status with every assembled section.**

# Assemble the current one-shot finding

| Field                     | Source-backed value                    |
|---------------------------|----------------------------------------|
| Confirmation sample       | 200,000 synthetic observations         |
| Selected region           | 12,575 cases; share 6.2875%            |
| Brier contrast            | 0.075524 against the complement        |
| Simultaneous 95% interval | [0.069631, 0.081417]                   |
| Selection and scope       | One-shot PAM; frozen models and region |

Attach the source record and its limitations to every value before drafting.

## Draft from the assembled evidence

“On 200,000 reserved synthetic observations, the frozen region contained 6.29% of cases. Its mean Brier loss exceeded the rest by 0.07552, with simultaneous 95% interval [0.06963, 0.08142]. These estimates are conditional on the fitted models and selected region.”

**Source: weighted-leaf one-shot confirmation record, 24 September 2026.**

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**Preserve metric, comparison, uncertainty and scope in the prose.**

# Represent missing evidence as an explicit gap

## ILLUSTRATIVE OTHER RECORD

Overall calibration: acceptable.

Exception: Segment B under stress.

Confirmation evidence: not supplied.

## REPORT CONTENT

Retain the overall assessment and its exception.

State that confirmation is unresolved.

Link each statement to the supplied record.

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**A missing confirmation record changes what can be concluded and must survive drafting.**

# Define the writing-prompt search

## INNER ARRAY

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Complete writing instructions built from declared factors.

Example factors: coverage, qualifications and citation placement.

## OUTER ARRAY

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Report cases with fixed evidence packets and required fields.

Include routine cases, exceptions and missing evidence.

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Score the writing procedure against the record and reporting requirements.

# Construct a small prompt design

Illustrative candidate instructions; the evidence packet remains fixed.

| Factor         | Level 0                | Level 1                    |
|----------------|------------------------|----------------------------|
| Coverage       | Section checklist      | Address each required item |
| Qualifications | Attach to each finding | Dedicated scope paragraph  |
| Citations      | Per factual sentence   | Per factual bullet         |

Three binary factors permit eight combinations before constraints; preserve mandatory evidence rules in each.

## Score drafts in separate dimensions

### FAITHFULNESS

Are claims, values and scope supported by the returned records?

Audit the claim extractor.

### COMPLETENESS

Are all required facts, exceptions and limitations represented?

Track missing item IDs.

### USABILITY

Are the conclusions clear, organized and appropriately qualified?

Use a validated assessment rubric.

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**Mandatory evidence checks govern eligibility; retain separate quality measures for comparison.**

# Refine instructions, then assess the frozen choice

## DISCOVERY

Compare candidate drafts on development packets.

Use failures to choose further factor settings.

Admit a new structure if needed.

## CONFIRMATION

Freeze instruction, builder and writing-model settings.

Compare with the baseline on reserved packets.

Record paired effects and uncertainty.

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**Keep reserved reporting cases outside both prompt selection and structural reflection.**

# Verify the report in both directions

## FROM TEXT TO EVIDENCE

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Check every asserted fact and number.  
Resolve its scope and source.  
Flag unsupported or contradicted claims.

## FROM REQUIREMENTS TO TEXT

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Check every required evidence item.  
Retain stress exceptions and limitations.  
Flag material omissions.

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Also compare the executive summary, detailed sections and tables for consistent conclusions.

# What the executed reporting replay establishes

## RECORDED RESULT

The regional notebook requires 25 fact IDs and six qualifications.

It accepts the complete plan and rejects two incomplete plans.

## EVIDENCE SCOPE

The assembled output contains all required IDs and qualifications.

This measures deterministic coverage checking and assembly.

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**LLM retrieval, full semantic drafting and the complete workflow need their own evaluation.**

# Deliver the report with a reviewable audit trail

## ASSESSMENT

Supported findings and materiality.

Scope and uncertainty.

Separate repair conclusions.

## TRACEABILITY

Source IDs, exact values, test versions and prompt lineage.

Checks and unresolved gaps.

## APPROVAL

Validator reviews interpretation and limitations.

Record the final decision and approved report version.

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**The report is complete when its claims, omissions, qualifications and approval can be inspected.**